

Heaven's Light is Our Guide



Department of Computer Science & Engineering
Rajshahi University of Engineering & Technology

Lab Manual

Course Title: Sessional based on CSE 4119

Course No.: CSE 4120

Credit: 0.75

Reference Book:

[1] Jiawei Han, Micheline Kamber, and Jian Pei, “Data Mining: Concepts and Techniques”

Experiments/Problems:

[Week 1] Introduction to Data Mining and Tools

Objective: Understand basic data mining concepts and tools.

Topics: Data types, tasks (classification, clustering, association).

Tools: Weka, [Python](#), or Orange.

Datasets:

- *Iris Dataset* – Classification, clustering
- *Online Retail* – Association rule mining (market basket) and Clustering
- *House Prices* – Regression

Tasks:

- Install Python libraries (e.g., Pandas, scikit-learn, Matplotlib).
- Explore datasets using pandas.

Lab Outcomes:

- Load dataset (in .csv or .xlsx format):

1. *Iris Dataset*

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa

2. *Online Retail*

	InvoiceNo	StockCode	...	CustomerID	Country
0	536365	85123A	...	17850.0	United Kingdom
1	536365	71053	...	17850.0	United Kingdom
2	536365	84406B	...	17850.0	United Kingdom
3	536365	84029G	...	17850.0	United Kingdom
4	536365	84029E	...	17850.0	United Kingdom

[5 rows x 8 columns]

3. *House Prices*

	price	area	bedrooms	...	parking	prefarea	furnishingstatus
0	13300000	7420	4	...	2	yes	furnished
1	12250000	8960	4	...	3	no	furnished
2	12250000	9960	3	...	2	yes	semi-furnished
3	12215000	7500	4	...	3	yes	furnished
4	11410000	7420	4	...	2	no	furnished

[5 rows x 13 columns]

- Try to understand the data type. Determine the dimensions of the data.
- Find out which one you can use as a label.
- Try different ways to split the data into training, validation, and test sets (60% training, 20% validation, and 20% test data).

- Identifying the dependent and independent variables.
- Split into two sets: features and labels.
- Identifying the number of labels and showing them. Plot them using Matplotlib.
- Based on the labels categorised, what type of problem is this: classification, multiclass classification, regression or clustering?